

Library Management System - Beginner Assignment

Functional Requirements

1. Book Management

- View all available books in the library
- Add new books to the library collection
- Update existing book information
- Remove books from the library
- Search books by title or author

2. Student Management

- Register new students in the system
- View all registered students
- Update student information
- Remove student records
- Search students by name

3. Book Borrowing System

- Allow students to borrow available books
- Track which books are currently borrowed
- Record the date when each book was borrowed
- Process book returns when students bring books back
- View borrowing history for each student
- Prevent borrowing if book is already checked out
- Display all currently borrowed books

4. Data Validation

- Ensure no duplicate book entries (by ISBN)
- Prevent deleting books that are currently borrowed
- Prevent deleting students with active borrowings
- Validate student contact information format

- Ensure book availability status is accurate

5. System Constraints

- A book can only be borrowed by one student at a time
- Students can borrow multiple books
- The system must maintain complete borrowing history
- All data changes must persist in the database

Technical Implementation Guidelines

- Use Spring Boot with Spring Data JPA
- Configure MySQL as the database
- Implement proper entity relationships
- Create necessary repository interfaces
- Follow RESTful API design principles
- Include proper error handling
- Use appropriate HTTP status codes